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Gut microbiome–derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease

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The effect of alterations in intestinal microbiota on microbial metabolites and on disease processes such as graft-versus-host disease (GVHD) is not known. Here we carried out an unbiased analysis to identify previously unidentified alterations in gastrointestinal microbiota–derived short-chain fatty acids (SCFAs) after allogeneic bone marrow transplant (allo-BMT). Alterations in the amount of only one SCFA, butyrate, were observed only in the intestinal tissue. The reduced butyrate in CD326$^+$ intestinal epithelial cells (IECs) after allo-BMT resulted in decreased histone acetylation, which was restored after local administration of exogenous butyrate. Butyrate restoration improved IEC junctional integrity, decreased apoptosis and mitigated GVHD. Furthermore, alteration of the indigenous microbiota with 17 rationally selected strains of high butyrate–producing Clostridia also decreased GVHD. These data demonstrate a heretofore unrecognized role of microbial metabolites and suggest that local and specific alteration of microbial metabolites has direct salutary effects on GVHD target tissues and can mitigate disease severity.

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Alterations in the intestinal microbiome are associated with several disease processes[1–5]. However, the effect that changes in the community structure of the microbiome have on the production of microbial-derived metabolites is poorly explored. Microbial metabolites influence disease severity, but whether alterations in microbial metabolites can affect outcomes after allogeneic bone marrow transplant (allo-BMT) is not known. Allo-BMT is a critical interventional therapy for patients with aggressive hematological malignancies[6,7]. Although allo-BMT is a curative and widely used treatment, approximately 40–50% of patients experience severe gastrointestinal damage from graft-versus-host disease (GVHD), which leads to high transplant-related mortality[7,8].

Studies have shown that the intestinal microbiota is considerably altered in patients with GVHD and that the alterations correlate with GVHD severity and pathogenesis[4,9]. Nevertheless, the direct causality between changes in the host microbiota and GVHD severity is unclear. Moreover, it remains unknown whether changes in the microbiota result in alterations in levels of microbial metabolites and by-products that have biological effects on allo-BMT recipients.

Microbial metabolites such as short-chain fatty acids (SCFAs) are exclusively derived from the gastrointestinal (GI) microbiota and are not made by the host. Some of these fatty acids (FAs), including the histone deacetylase inhibitor (HDACi) butyrate, are a preferred energy source for intestinal epithelial cells[10–13] (IECs), and administration of exogenous HDACi regulates GVHD[14–16]. But the effect that host indigenous microbial metabolites that function as HDACis have on GVHD remains unknown[11–13].

Here we performed unbiased profiling of the microbial metabolome with a specific focus on targeted FAs after experimental allo-BMT. We found that the amount of only one SCFA, butyrate, was significantly decreased only in the intestinal tissue of allo-BMT recipients, resulting in decreased acetylation of histone H4 in IECs. Increasing the amount of intestinal butyrate restored acetylation of histone H4, protected IECs and decreased the severity of GVHD. Furthermore, rationally altering host GI microbiota to high-butyrate producers[17] mitigated GVHD.

RESULTS

Targeted microbial-metabolite profiling

We hypothesized that alterations in the composition of the microbiota in the GI lumen would result in an altered microbial metabolome after the onset of GVHD[4,18]. We determined the concentration of microbial

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FA metabolites, both SCFAs and long-chain FAs up to 18 carbons in length, from several sites in mice 7 d after bone marrow transplant (BMT) (day +7). We analyzed the serum, spleen, liver, intestines and luminal contents (stool) of the intestines using gas chromatography combined with mass spectrometry (GC-MS) (**Supplementary Fig. 1a**). We used a well-established, clinically relevant model of major histocompatibility complex (MHC)-mismatched BMT in which cells from C57BL/6J (H-2^b) mice are transferred to lethally irradiated BALB/c (H-2^d) mice and compared it to syngeneic transplant and naive animals. The animal cages were exchanged between groups on day +3 to account for any alterations in the microbial environment, and analysis after GC-MS was performed in a blinded manner. The FA concentrations in the luminal contents of the intestines were not significantly different between any of the groups (**Fig. 1a** and **Supplementary Fig. 1b**). We also found no significant differences in FA concentrations in the serum or in tissues such as the spleen and liver between allogeneic animals (**Supplementary Fig. 1c** and **Supplementary Table 1**) and syngeneic animals or naive controls. However, the greatest—and only statistically significant—difference was observed in the SCFA butyrate, which was significantly decreased in amount only in the intestinal tissue at day 7 (**Fig. 1b** and **Supplementary Fig. 1d**). We once again observed similar results on day 21 as on day 7 (**Supplementary Fig. 2**). Collectively these data demonstrate that butyrate levels are consistently decreased only in the intestinal tissue after allo-BMT.

Functional impact of altered levels of SCFAs in IECs

In light of the decreased amounts of butyrate in allogeneic animals only in the intestinal tissue, we next analyzed the potential functional impact of decreased butyrate in IECs. Because butyrate is an HDACi[10,12,19], we examined the degree of histone acetylation by immunoblotting purified CD326$^+$ IECs collected from mice after BMT. The degree of acetylation of histone H4 was significantly decreased on day 7 ($P < 0.0025$; **Supplementary Fig. 3a**) and day 21 (**Fig. 2a**) after allo-BMT, demonstrating that reductions in the amount of butyrate resulted in decreased histone acetylation. Therefore, to confirm that the decreased acetylation observed after allo-BMT (compared to syngeneic controls) was secondary to decreased HDAC inhibition resulting from decreased amounts of butyrate and not to potential alterations in HDAC and histone acetyltransferase (HAT) enzyme levels[20] after transplant, we analyzed the expression of HDACs and HATs in IECs after BMT. We observed similar levels of several HDACs (*Hdac1*, *-4*, *-7*, *-9* and *-10*) (**Fig. 2b**) and HATs (*Ep300* and *Kat5*) (**Fig. 2c**) by qPCR in the IECs (CD326$^+$) of syngeneic and allogeneic BMT recipients. Furthermore, HDAC (**Fig. 2d**) and HAT (**Fig. 2e**) enzyme activity did not differ between these groups of animals. These data suggest that the reduction in histone acetylation in IECs after allo-BMT is a result of decreased amounts of butyrate.

Decreased uptake of butyrate by IECs

We next explored whether the diminished concentration of butyrate observed in the intestinal tissue was from impaired uptake of butyrate after allo-BMT. To this end, we examined the expression of the known butyrate monocarboxylate transporter SLC5A8 and of the butyrate receptor GPR43 in IECs after allo-BMT[10,21]. Decreased gene (**Fig. 2f**) and protein (**Fig. 2g**) expression of SLC5A8 and GPR43 (**Supplementary Fig. 3b,c**) were observed in IECs from allogeneic animals compared with those from syngeneic animals after transplant on both day +21 and day +7 (**Supplementary Fig. 3d**), which suggested that the decreased butyrate concentration in the IECs was due to decreased uptake of microbiota-derived luminal butyrate. We next cultured primary IECs with pro-inflammatory mediators (interferon-γ and/or tumor necrosis factor) and analyzed the expression of SLC5A8. Exposure of IECs to inflammatory cytokines significantly decreased the expression of *Slc5a8* compared to that in vehicle-treated controls (**Supplementary Fig. 3e**). These data indicate that the intense inflammatory milieu after allo-BMT causes decreased expression of butyrate transporters and receptors, leading to decreased butyrate and histone acetylation in IECs.

Rescuing the cellular effects of decreased butyrate

We next determined whether the decreased amount of butyrate in IECs could be restored *in vivo* and, further, whether this would have a functional effect on histone acetylation. In addition to using transporters, butyrate can diffuse across the mucosal barrier into IECs when present in high concentrations[10,22]. Therefore, we hypothesized that local administration of high amounts of butyrate would restore histone acetylation in IECs *in vivo*. To test this, we transferred cells from C57BL/6J mice into BALB/c mice and administered vehicle or butyrate via daily intragastric gavage. Daily butyrate administration

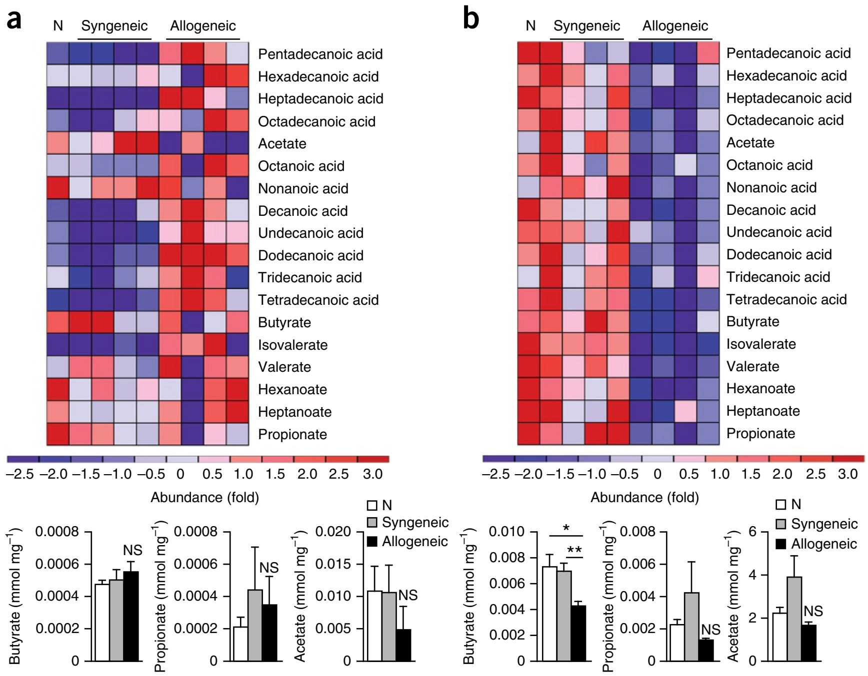

Figure 1 Allo-BMT reduces amounts of intracellular butyrate in IECs. (**a**,**b**) Abundance of fatty acids (short and long chain) on day +7 in the intestinal luminal contents (stool) (**a**) and intestinal tissue (**b**) of recipients of syngeneic BMT (BALB/c→BALB/c), allo-BMT (C57BL/6J→BALB/c) or no BMT (naive (N)). Graphical results for all animals combined are shown below the heat maps for SCFAs butyrate, propionate and acetate. Representative heat maps are shown for $n = 10$ mice in the naive and syngeneic groups and $n = 9$ mice in the allogeneic group. $*P < 0.05$, $**P < 0.01$, analysis of variance (ANOVA). Bars and error bars represent mean and s.e.m., respectively.

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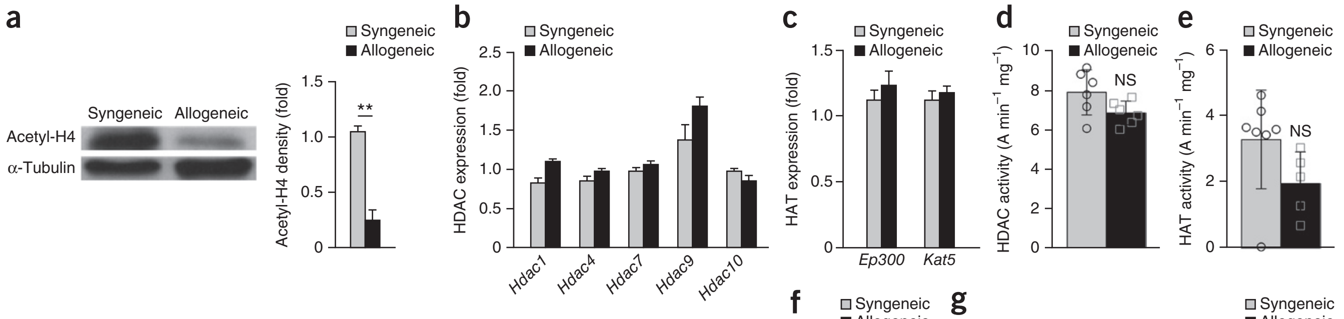

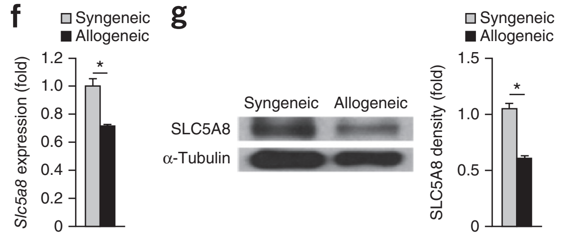

Figure 2 Loss of butyrate decreases histone acetylation in IECs. (**a**) Expression of acetylated histone H4 (top blot) and densitometric analysis normalized to the presence of α-tubulin (bottom) 21 d after syngeneic (BALB/c→BALB/c) or allogeneic (C57BL/6J→BALB/c) BMT. Results are representative of three experiments combined, each with $n = 5$–6 mice per group. (**b**–**e**) Gene expression of representatives of class I, II and IV HDAC enzymes (**b**), HAT enzyme expression (**c**), HDAC activity (**d**) and HAT activity (**e**) in CD326$^+$ IECs from recipients of syngeneic or allogeneic BMT. A, absorbance. In **d** and **e**, open circles represent individual syngeneic mice, and open squares represent individual allogeneic mice. (**f**,**g**) Gene expression (**f**) and protein density (**g**) of SLC5A8 (monocarboxylate transporter of butyrate) in IECs (CD326$^+$) of syngeneic and allogeneic transplant recipients 21 d after BMT. Representative immunoblots; data are representative of three similar experiments. $*P < 0.05$, $**P < 0.01$, Student's *t*-test. NS, not significant. Bars and error bars represent the mean and s.e.m., respectively.

for 21 d significantly restored acetylation of histone H4 compared with no treatment in allo-BMT recipients (**Fig. 3a**).

We next determined whether there was a difference in the uptake and metabolism of exogenously administered butyrate between recipients of syngeneic versus allogeneic BMT. To this end we performed metabolic flux analysis to assess the incorporation of heavy ^{13}C–labeled butyrate into luminal and intestinal tissue butyrate pools by using GC-MS as described above. We treated recipients of syngeneic or allogeneic transplant 7 d after BMT with a bolus of either ^{13}C-butyrate or ^{12}C-butyrate, which served as the control. We collected IECs and luminal contents from the mice 6 h later and analyzed them for incorporation of ^{13}C. Although we saw similar amounts of heavy ^{13}C–labeled butyrate in the lumen of the large intestine, we saw significantly less ^{13}C-labeled butyrate in the intestinal tissue of recipients of allo-BMT, suggesting decreased uptake (**Fig. 3b**). To determine whether there were any differences in butyrate metabolism, we further examined the presence of ^{13}C-butyrate at different stages of the tricarboxylic acid (TCA) cycle. Specifically, we determined the incorporation of ^{13}C-butyrate into citrate, succinate and, further downstream, malate (**Supplementary Fig. 3f**). There was a significant difference between syngeneic BMT and allo-BMT recipients in the amount of heavy carbon in malate (**Fig. 3c**), as well as a trend toward greater label incorporation into citrate (**Fig. 3c**) and succinate (**Supplementary Fig. 3g**) in IECs from the large intestine of allo-BMT recipients. Similarly, examination of IECs from both small and large intestines showed an overall increased level of ^{13}C incorporation in the downstream metabolite of the TCA cycle, malate, in allo-BMT recipients compared with syngeneic BMT recipients (**Supplementary Fig. 3h**), suggesting an increased rate of metabolism in the IECs of those animals.

Furthermore, daily intragastric gavage of butyrate resulted in increased amounts of the butyrate transporter SLC5A8 compared with those in untreated animals (**Fig. 3d**), suggesting that butyrate has a positive feedback mechanism that results in an increase in the amount of its transporter. To determine whether butyrate was directly responsible for induction of its own transporter, we analyzed the degree of histone acetylation at the promoter of SLC5A8 with chromatin immunoprecipitation (ChIP). We found an increased association of acetylated histone H4 in the promoter region of *Slc5a8* in IECs (CD326$^+$) treated with butyrate compared with vehicle-treated cells (**Fig. 3e**). These data demonstrate that the decreased amount of butyrate after allo-BMT has functional effects on IECs.

Increased amounts of intestinal butyrate mitigated GVHD

Systemic administration of HDACi decreases the severity of acute GVHD[14,15,23]. We therefore determined whether increasing local amounts of the endogenous HDACi butyrate would affect GVHD severity. Again using the C57BL/6J-into-BALB/c model, we administered vehicle control or butyrate via intragastric gavage daily for 1 week and then every other day for the remainder of the experiment.

Administration of butyrate resulted in decreased weight loss (**Fig. 3f**), decreased GVHD clinical scores (**Fig. 3g**) and increased survival (**Fig. 3h**) compared with vehicle treatment. We observed similarly improved survival using a second clinical model of BMT, an MHC-matched, minor antigen–mismatched model. We transferred C3H.SW (H-2^b) cells into C57BL/6J (H-2^b) mice, thus demonstrating strain-independent results (**Supplementary Fig. 4a**). Furthermore, histopathological analysis 21 d after BMT showed decreased histological scores in the intestines of the major MHC–mismatch model (**Fig. 3i**) and decreased histological scores in both intestines and liver in the model of minor antigen mismatch (**Supplementary Fig. 4b**). To determine whether irradiation-related inflammation was critical for butyrate-induced protection, we next examined the butyrate-induced protective effects in a non-irradiated model of BMT from parent (C57BL/6J, H-2^b) into F1 (B6D2F1, H-2$^{b/d}$). We once again observed significantly less weight loss ($P < 0.011$; **Supplementary Fig. 4c**) and improved survival (**Supplementary Fig. 4d**) in allogeneic recipients that were treated with intragastric butyrate compared with vehicle-treated controls. Next, to further evaluate the magnitude of the butyrate-induced protective effect, we determined GVHD mortality, once again using the MHC-disparate B6-into-BALB/c model. Intragastric gavage of butyrate induced significant GVHD survival benefit in allogeneic animals that received higher doses of T cells ($P < 0.0001$; **Supplementary Fig. 4e**). These data collectively show that butyrate induced GVHD protective effects regardless of strain combinations, conditioning or higher alloreactive T cell doses.

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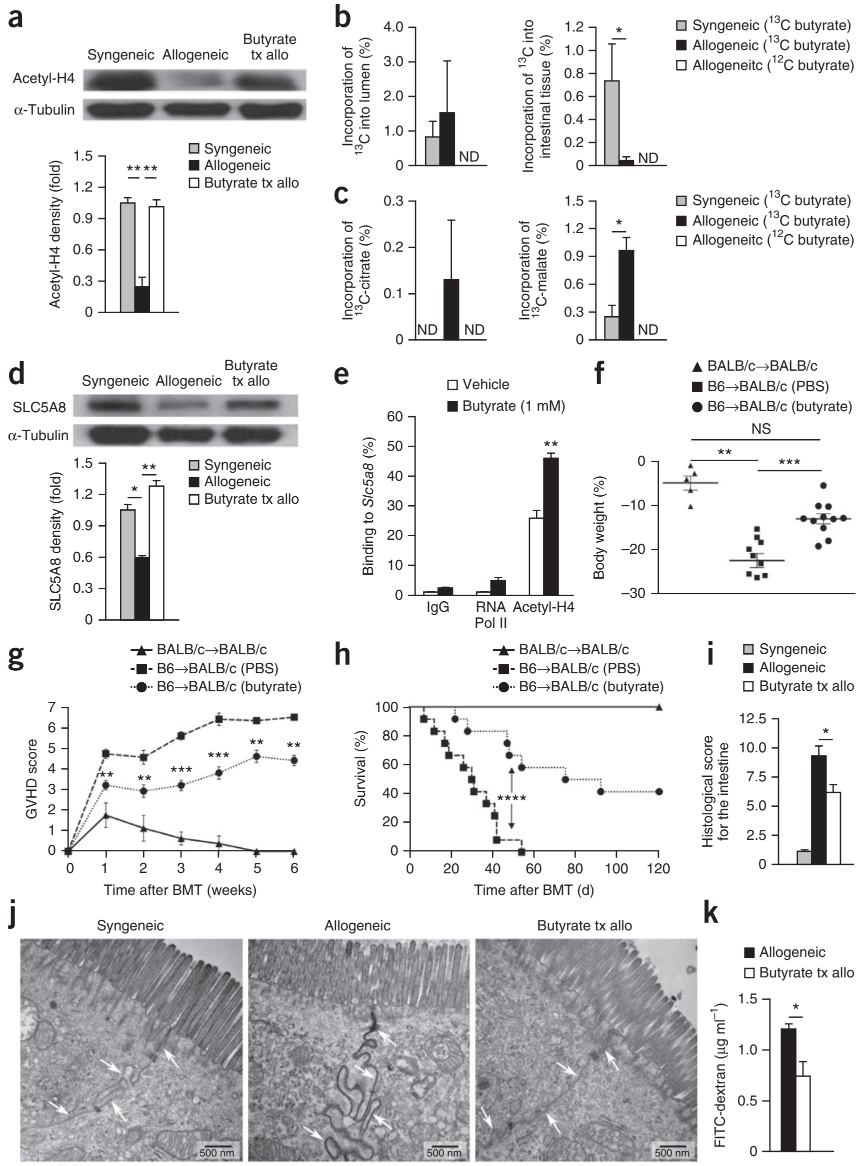

Figure 3 Intragastric gavage with butyrate enhances uptake of the SCFA and histone acetylation in IECs. (**a**) Representative immunoblots of CD326+ IECs from syngeneic (BALB/c→BALB/c) and allogeneic (C57BL/6J→ BALB/c) BMT recipients that received vehicle or butyrate (10 mg kg−1; "Butyrate tx allo"). The graph shows amounts of acetylated histone H4 (acetyl-H4) 21 d after BMT; data are representative of three experiments combined. (**b**) Incorporation of ^{13}C-labeled butyrate in luminal contents and intestinal tissue of the large intestine in mice gavaged with ^{13}C-butyrate or (non-labeled) ^{12}C-butyrate; $n = 6$ mice per group. ND, not detected. (**c**) ^{13}C label incorporation in tricarboxylic acid metabolite pools from the large intestines of mice fed a bolus (2 g kg−1) of ^{13}C-butyrate or ^{12}C-butyrate; $n = 6$ mice per group. (**d**) SLC5A8 in IECs of syngeneic and allogeneic BMT recipients 21 d after transplant (as in **a**). (**e**) ChIP of butyrate-treated IECs (CD326+) and binding of acetylated histone H4 in the promoter region of *Slc5a8*. IgG, immunoglobulin G; Pol, polymerase. (**f**–**i**) Weight loss on day 21 (**f**), GVHD clinical score (**g**), survival (**h**) and intestinal histopathology 21 d after BMT for recipients treated with intragastric vehicle or butyrate (**i**); syngeneic, $n = 6$ mice per group; allogeneic, $n = 12$ mice per group. In **f**, symbols represent individual data points, and horizontal bars represent means. In **g** and **h**, symbols represent mean values. (**j**) Transmission electron microscopy images of intestines isolated from BMT recipients with or without intragastric gavage of butyrate for the duration of the experiment. All samples were isolated 7 d after BMT and stained with ruthenium red (0.1%); arrows indicate cell-cell interfaces. (**k**) FITC-dextran translocation across the GI barrier into blood serum in butyrate-treated allo-BMT recipients compared to vehicle-treated all-BMT recipients 21 d after BMT. B6, C57BL/6J. $*P < 0.05$, $**P < 0.01$, $***P < 0.0001$, $****P < 0.00001$, ANOVA (**a**–**d**,**i**,**k**), Student's *t*-test (**e**–**g**) or Mantel-Cox log-rank test (**h**). Bars and error bars represent the mean and s.e.m., respectively.

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Increased amounts of intracellular butyrate protected GI epithelium

We next determined whether the decreased severity of GI GVHD resulted in decreased translocation of luminal contents and improved epithelial integrity[24–26]. We used transmission electron microscopy to examine the ability of butyrate to preserve cellular junctions after allo-BMT. We found intense leakage of the electron-dense stain ruthenium red[25] in allo-BMT recipients treated with vehicle alone (**Fig. 3j**). However, the integrity of the IEC junction was preserved at both day 7 (**Fig. 3j**) and day 21 (**Supplementary Fig. 4f**) in allo-BMT recipients that received local intragastric administration of butyrate.

We further assessed the intestinal permeability after allo-BMT by intragastric administration of fluorescein isothiocyanate (FITC)-dextran, a non-metabolized carbohydrate[27]. Butyrate-treated allo-BMT recipients exhibited significantly less detectable FITC-dextran in the serum at 21 d after transplant (**Fig. 3k**).

Reduced severity of GVHD is independent of donor T_{reg} cells

Regulatory T (T_{reg}) cells mitigate GVHD, and butyrate has been shown to increase the number of intestinal T_{reg} cells[17,28]. We therefore analyzed the cellular contents of the intestine 21 d after allo-BMT to determine whether butyrate had an effect on local T_{reg} cells. The total numbers of CD45.1+ cells recovered from the intestinal lamina propria were not different between vehicle- and butyrate-treated allo-BMT recipients (**Supplementary Fig. 4g**). In contrast, intestinal infiltration of donor CD4+ and CD8+ T cells and activated T cells (CD69+ or CD44hi) was decreased in animals that received local intragastric butyrate administration (**Fig. 4a**). However, the ratio of donor T_{reg} cells to effector T cells was not different in the intestines of those animals (**Fig. 4b**). Microbiota-derived butyrate has been shown to increase the number of immune-regulatory macrophages in the GI tract, which increases the number of T_{reg} cells[29]. However, the total number of donor macrophages in the intestine of allogenenic recipients did not differ between vehicle- and butyrate-treated mice in our experiment (**Fig. 4c**).

To further determine whether the salutary effects of local treatment with butyrate on GI GVHD were dependent on donor T_{reg} cells, we next performed a BMT using the same MHC-mismatched BMT model in which cells from C57BL/6J (H-2b) mice were transferred to BALB/c (H-2d) mice. We transferred T cell–depleted (TCD) bone marrow and purified CD4+CD25− T cells from donors and administered vehicle or

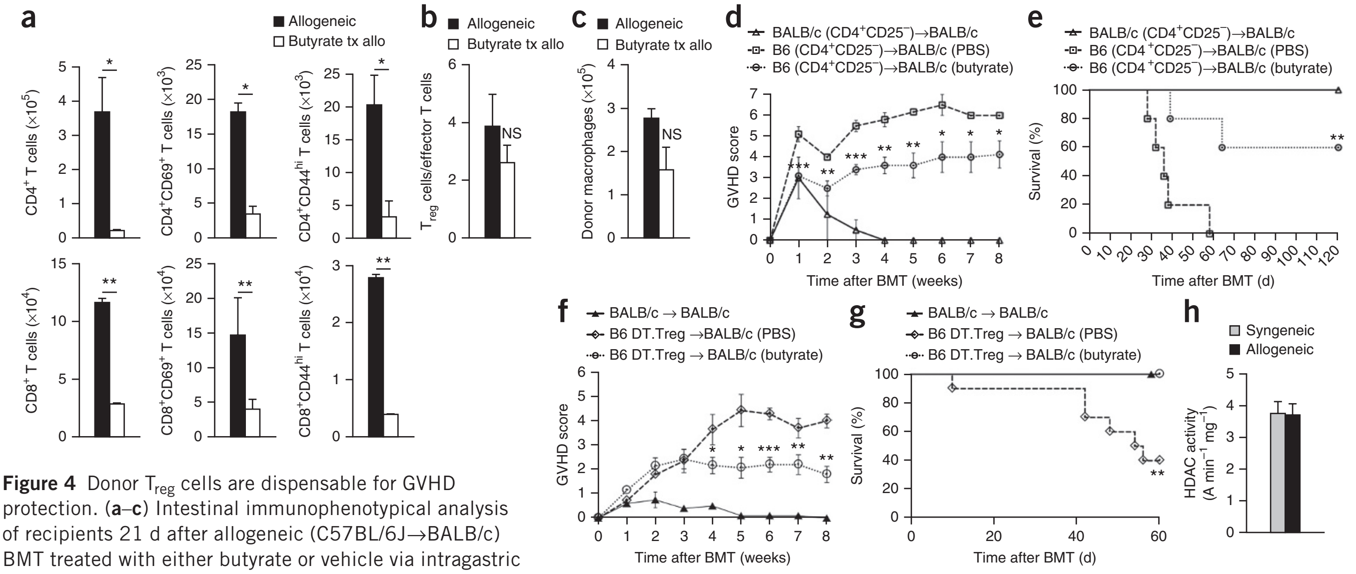

Figure 4 Donor T_{reg} cells are dispensable for GVHD protection. (**a**–**c**) Intestinal immunophenotypical analysis of recipients 21 d after allogeneic (C57BL/6J→BALB/c) BMT treated with either butyrate or vehicle via intragastric gavage. (**a**) Total numbers of intestinal CD4+ and CD8+ T cells (left) and of activated CD69+ T cells (middle) and CD44^{hi} T cells (right). (**b**,**c**) Ratio of intestinal T_{reg} cells (CD4+CD25+Foxp3+) to effector T cells (CD4+Foxp3−) (**b**) and total number of intestinal donor macrophages (CD11b+F4.80+) (**c**) in recipients of allo-BMT treated with vehicle or butyrate. NS, not significant. (**d**,**e**) Data for recipients of 1.5×10^6 CD4+CD25− (T_{reg}-depleted) donor T cells and TCD bone marrow. GVHD score (**d**) and survival (**e**) in BMT recipients treated with vehicle or butyrate (10 mg kg^{-1}) or not treated; $n = 6$ (syngeneic) or 12 (allogeneic) mice per group. (**f**,**g**) Data for recipients of 0.5×10^6 T_{reg} cell–depleted (C57BL/6J DT.T_{reg}) donor cells and TCD bone marrow: (**f**) GVHD score and (**g**) survival in BMT recipients treated with PBS vehicle or butyrate or not treated as indicated. In **d**–**g**, symbols represent mean values and significance is determined between butyrate-treated and PBS-treated allo-BMT mice. (**h**) HDAC activity in splenic T cells 21 d after syngeneic or allogeneic BMT. $n = 5$ (syngeneic) or 10 (allogeneic) mice per group. $*P < 0.05$, $**P < 0.01$, $***P < 0.0001$, Student's *t*-test (**a**–**d**,**f**) or Mantel-Cox log-rank test (**e**,**g**). Bars and error bars represent the means and s.e.m., respectively.

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butyrate to the recipients daily for 1 week and every other day thereafter. We still observed decreased GVHD severity (**Fig. 4d**,**e**) indicating that donor T_{reg} cells may be dispensable for the mitigation of GVHD.

To further confirm the donor T_{reg} cell–independent protective effects of butyrate on GVHD, we next used donor C57BL/6J mice with a knock-in of human diphtheria toxin receptor expressed only on T_{reg} cells (DREG mice)[30,31]. We injected DREG donor mice with diphtheria toxin (10 μg kg^{-1}) on day −2 and day −1 and confirmed loss of Foxp3− T_{reg} cell expression (**Supplementary Fig. 4h**). The T_{reg} cell–depleted T cells were then used as donor cells in the major MHC–mismatch model described above. We again saw significantly improved clinical GVHD (**Fig. 4f**), less weight loss (**Supplementary Fig. 4i**) and improved survival (**Fig. 4g**) compared with PBS-treated controls.

To further examine whether local administration of butyrate affects donor T cells directly, we determined the HDAC and HAT enzymatic activity in donor T cells collected from the recipient mice. Both the HDAC activity (**Fig. 4h**) and the HAT activity (**Supplementary Fig. 4j**) in donor T cells collected from spleens of recipient mice were similar between butyrate-treated and vehicle-treated allogeneic animals. These data collectively demonstrate that the reduction in GI GVHD severity after intragastric administration of butyrate is independent of potential effects from donor T_{reg} cells.

Butyrate protects IECs from alloreactive T cell–mediated damage

We next sought to explore the potential mechanisms that contribute to butyrate-induced protection from severe GVHD. Because (a) butyrate was present in decreased amounts in IECs and (b) administration of butyrate mitigated GI GVHD independently of T_{reg} cells but (c) improved junction integrity, we explored whether butyrate had direct effects in the protection of IECs from alloreactive T cell–mediated damage and conditioning. We treated IECs *ex vivo* with vehicle or butyrate for 24 h and subjected the cells to either 6-Gy irradiation or no irradiation, followed by 24 h of additional incubation with butyrate. We observed that butyrate was not toxic to IECs and conferred protection from irradiation-induced apoptosis (**Fig. 5a**).

We next determined the ability of butyrate-treated IECs to withstand damage mediated by alloreactive T cells. We isolated and cultured primary IECs with butyrate or vehicle control overnight. The pretreated IECs were next cocultured with primed allogeneic CD8+ T cells in the absence of butyrate. Fewer butyrate-pretreated IECs succumbed to CD8+ T cell killing within 6 h and 16 h after coculture compared with controls (**Fig. 5b**).

Because butyrate is a primary energy source for IECs[11–13], we next determined whether butyrate enhances the survival and growth of IECs *in vitro*. To this end, we cultured intestinal organoids in the presence or absence of butyrate. We observed that culture in the presence of butyrate significantly increased organoid size compared with untreated controls (**Fig. 5c**). Next we confirmed the effect of butyrate on IEC junctional function in the organoid cultures by determining the mRNA expression of claudins. We observed that culture of organoids with butyrate significantly increased the mRNA expression of claudins (*Cldn1*, *Cldn5*, *Cldn6*, *Cldn10*, *Cldn11*, *Cldn13*, *Cldn14*, *Cldn17* and *Cldn18*) ($P < 0.0113$; **Supplementary Fig. 5a**).

Molecular mechanisms of IEC protection

We hypothesized that butyrate would increase the expression of anti-apoptotic genes via modulation of histone acetylation. We analyzed pro- and anti-apoptotic mRNA expression levels[32] and found that expression of *Bak1* and *Bax* was significantly decreased ($P < 0.0012$; **Supplementary Fig. 5b**), whereas amounts of transcripts of the anti-apoptotic protein BCL-B (*Bcl2l10*) were significantly increased (**Fig. 5d**), in butyrate-treated IECs versus untreated controls. We next examined

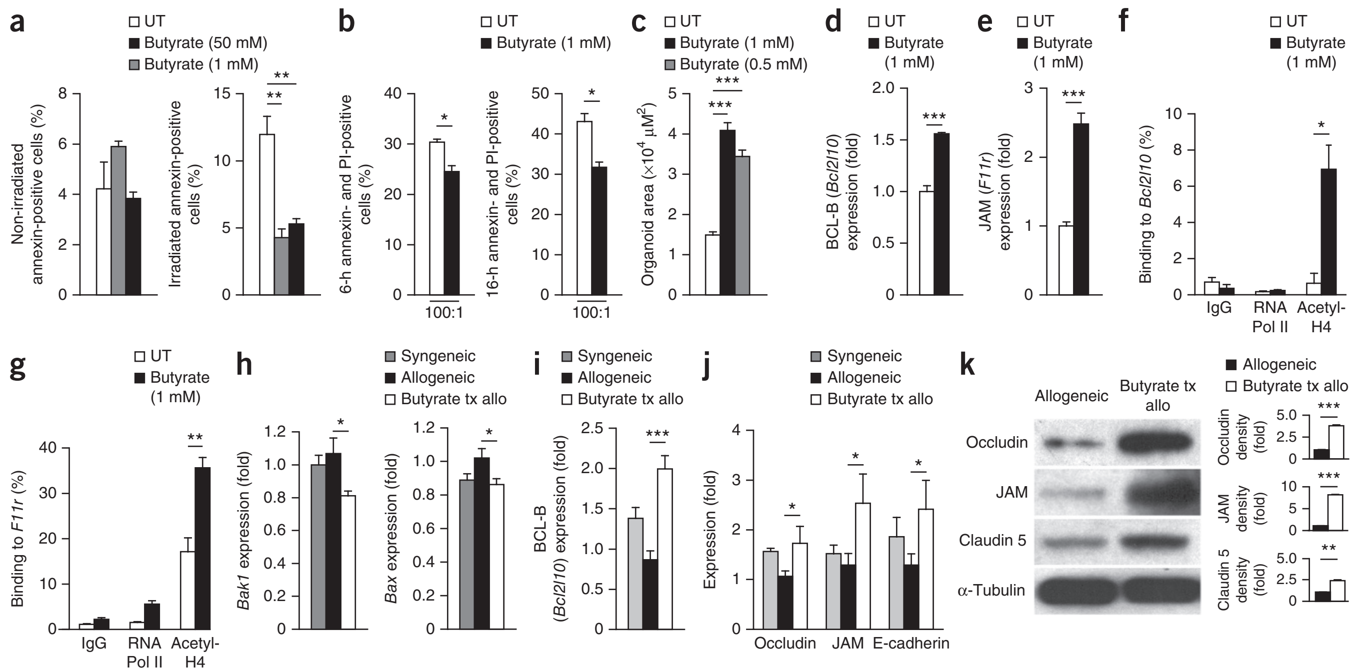

Figure 5 Butyrate treatment enhances the IEC barrier. (**a**) Number of annexin-positive CD326+ purified IECs after culture in the presence or absence of butyrate and irradiation (6 Gy) (right) or no irradiation (left). UT, no butyrate treatment. (**b**) Allogeneic CD8+ T cell killing assay. Graphs show numbers of CD326+ IECs positive for annexin and propidium iodide (PI) after overnight incubation with or without butyrate followed by coculture with alloreactivity-primed CD8+ T cells. (**c**) Area of cultured primary intestinal organoids after butyrate treatment compared to untreated controls. (**d**,**e**) Expression of (**d**) anti-apoptotic protein BCL-B (*Bcl2l10*) and (**e**) junctional protein JAM (*F11r*) in primary CD326+ IECs in the presence of 1 mM butyrate compared to untreated controls. (**f**,**g**) ChIP assay of butyrate-treated IECs (CD326+) binding acetylated histone H4 in the promoter regions of (**f**) *Bcl2l10* and (**g**) *F11r*. (**h**–**j**) Analysis of IECs (CD326+) isolated from recipients of syngeneic (BALB/c→BALB/c) and allogeneic (C57BL/6J→BALB/c) BMT treated with butyrate or vehicle daily via intragastric gavage for 21 d. (**h**) Expression of pro-apoptotic genes *Bak1* (left) and *Bax* (right), (**i**) anti-apoptotic protein BCL-B (*Bcl2l10*) and (**j**) junctional proteins 21 d after BMT. (**k**) Representative immunoblot of CD326+ purified IECs, isolated 21 d after BMT, from allogeneic (C57BL/6J→BALB/c) BMT recipients treated daily with intragastric vehicle or butyrate (10 mg kg^{-1}). Graphs at right show densitometric analysis of two experiments combined; black bars represent samples from PBS-treated allo-BMT mice, and white bars samples from butyrate-treated allo-BMT mice. $n = 5$ (syngeneic) or 10 (allogeneic) mice per group. *$P < 0.05$, **$P < 0.01$, ***$P < 0.0001$, Student's *t*-test (**a**–**g**,**k**) or ANOVA (**h**–**j**). Bars and error bars represent the means and s.e.m., respectively.

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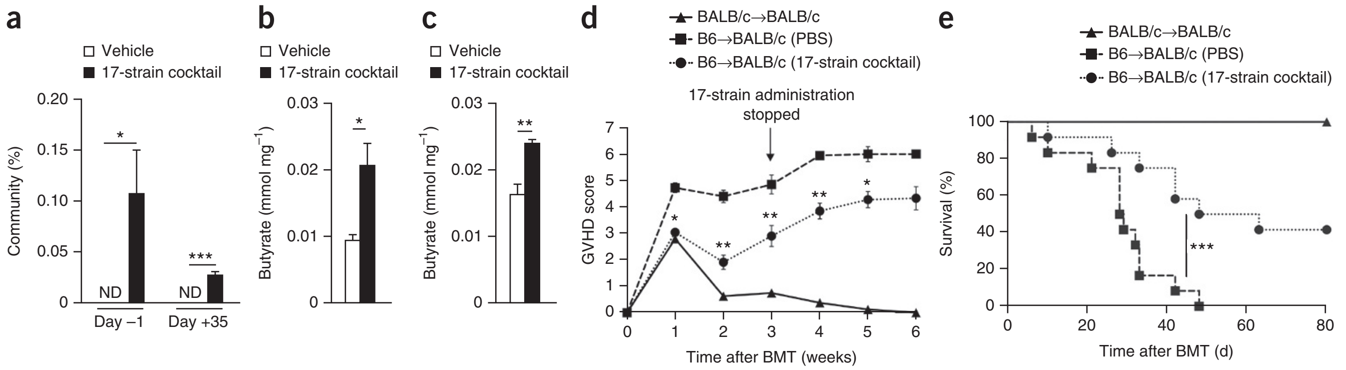
mRNA expression of junctional proteins such as occludin (*Ocln*) ($P < 0.04$; **Supplementary Fig. 5c**) and JAM (*F11r*) (**Fig. 5e**) in IECs after butyrate treatment, which significantly increased their expression. We also used ChIP to determine whether the restored acetylation of histone H4 observed in butyrate-treated allo-BMT recipients was responsible for increased BCL-B (*Bcl2l10*) and JAM (*F11r*) expression. We found that acetylation of histone H4 was indeed associated with the promoter regions of *Bcl2l10* (**Fig. 5f**) and *F11r* (**Fig. 5g**) in butyrate-treated IECs (CD326+).

These data thus collectively suggest that butyrate has several salutary effects on IECs that may or may not be mutually exclusive, such as regulating the expression of genes involved in reducing IEC apoptosis and increasing expression of junctional proteins in IECs. To determine whether these factors are involved in *in vivo* protection from GVHD, we determined the expression of pro- and anti-apoptotic proteins as well as junctional proteins in IECs isolated 21 d after allo-BMT. Amounts of pro-apoptotic transcripts of *Bak1* and *Bax* were significantly decreased in allo-BMT recipients that received intragastric butyrate treatment (**Fig. 5h**), whereas expression of the anti-apoptotic BCL-B (*Bcl2l10*) was increased (**Fig. 5i**). Further, expression of junctional proteins was similarly increased in butyrate-treated animals (**Fig. 5j**). To determine whether these results have biological consequences for protein expression, we also examined the amounts of occludin, JAM and claudin 5. Indeed, immunoblot analysis showed increased amounts of junctional proteins in recipients of allo-BMT treated with intragastric butyrate compared with their untreated counterparts (**Fig. 5k**). Overall, our results identify several ways in which butyrate can directly enhance epithelial cell function, ranging from protection from irradiation and alloreactive T cell–mediated apoptosis to proliferation and junctional protein expression, both *in vitro* and *in vivo*.

High butyrate–producing microbiota mitigate GVHD

The endogenous HDACi butyrate is a by-product of microbial fermentation[33]. Therefore, we next tested the hypothesis that altering the composition of indigenous GI microbiota in hosts to those that can produce high levels of butyrate will mitigate GVHD. We used 17 rationally selected strains of Clostridia that have been shown to increase amounts of butyrate both *in vitro* and *in vivo*[17,34]. We administered these strains via intragastric gavage every other day to naive mice starting 14 d before allo-BMT and continuing for 21 d after BMT. We characterized the microbiota in feces collected from animals that received vehicle and 17-strain administration by 16S rRNA-encoding gene sequencing. For animals that received the 17 Clostridial strains, 16S analysis[35–37] showed an important biologically significant shift in the microbiota, indicating that the introduced organisms could be detected (**Fig. 6a** and **Supplementary Fig. 6a**). Furthermore, GC-MS analysis 21 d after allo-BMT showed a significant increase in the amount of butyrate in the luminal contents (**Fig. 6b**) and in the intestinal tissues (**Fig. 6c**) of animals that received intragastric



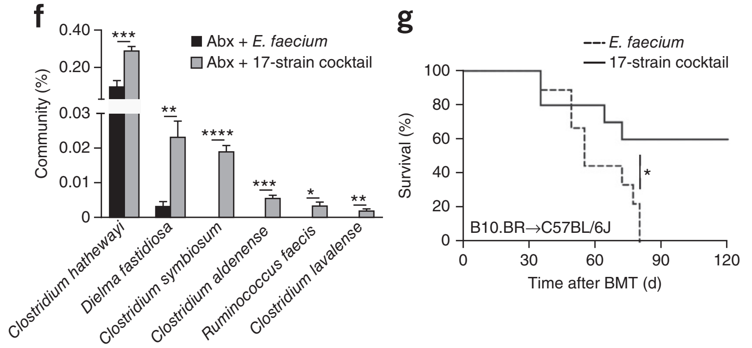


Figure 6 Rationally altering the commensal microbiota reduces GVHD severity. (**a**) The percentage of the total GI community made up of 17 specified Clostridial strains on day −1 and day +35 in vehicle-treated versus Clostridia-treated allo-BMT (C57BL/6J→BALB/c) recipients, as determined by 16S rRNA-encoding gene sequencing of stool. Vehicle or 17-strain cocktail was administered to recipients every other day via intragastric gavage beginning on day −14 and continuing through day +21. ND, not detected. (**b**,**c**) GC-MS analysis of intestinal luminal contents (stool) (**b**) and intestinal tissue (**c**) collected from recipients of allo-BMT gavaged with vehicle or colonized with 17-strain cocktail and killed 21 d after BMT. (**d**,**e**) GVHD clinical score (**d**) and survival (**e**) after syngeneic (BALB/c→BALB/c) and allogeneic (C57BL/6J →BALB/c) BMT with 17-strain administration compared to vehicle (PBS) control; $n = 6$ (syngeneic) or 12 (allogeneic) mice per group. In **d** and **e**, symbols represent mean values and significance is determined between Clostridia-treated and PBS-treated allo-BMT mice. (**f**,**g**) GI community analysis (**f**) and survival (**g**) of allo-BMT (B10.BR → C57BL/6J) recipients treated as indicated. (**f**) 16S gene sequencing of stool collected on day −1. Abx, antibiotic cocktail (defined in the text). (**g**) Survival of allo-BMT recipients treated with intragastric gavage of 17–Clostridial strain cocktail ($n = 10$) or *E. faecium* ($n = 9$). $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.0001$, $^{****}P < 0.00001$, Student's *t*-test (**a**–**d**,**f**) or Mantel-Cox log-rank test (**e**,**g**). Bars and error bars represent the means and s.e.m., respectively.

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gavage of the 17 Clostridial strains. The recipients of intragastric gavage of the 17 strains and allo-BMT exhibited significantly decreased GVHD severity (**Fig. 6d**,**e**). Detectable levels of the 17 Clostridial strains were diminished within 2 weeks (day +35) of cessation of intragastric administration (**Fig. 6a**).

Because microbiota variations are known to occur in different colonies of mice, we also determined whether mice housed at another institution (Memorial Sloan Kettering Cancer Center) and treated with the same 17 Clostridial strains also showed mitigated GVHD. Additionally, because clinical BMT patients are often treated with antibiotics, and as an alternative approach to colonizing the indigenous microbiota, we treated C57BL/6J mice with an antibiotic cocktail (5 mg of ampicillin, 4 mg of metronidazole, 5 mg of clindamycin and 5 mg of vancomycin) daily by intragastric gavage for 6 d to target obligate anaerobes. The mice were then colonized 4 and 6 d later with either human *Enterococcus faecium* or the same cocktail of 17 strains of human Clostridia[17] by intragastric gavage. Once again, we characterized the fecal microbiota by 16S gene sequence analysis on day −1 relative to BMT. We observed increased presence of *Clostridia* species in recipients of the cocktail of 17 Clostridial strains (**Fig. 6f**) compared with recipients of *E. faecium*. The mice were next used as recipients of an MHC-mismatched B10.BR (H-2^k) BMT and followed for survival. Animals that were treated with antibiotics but were not recolonized with bacteria died significantly faster ($P < 0.0001$) than mice not treated with the antibiotic mixture (**Supplementary Fig. 6b**). These data demonstrate that antibiotic treatment eliminated beneficial microbiota, similar to a previous report[6]. We once again observed significantly increased survival in the animals treated with the cocktail of 17 Clostridial strains (**Fig. 6g**). These results suggest that altering the indigenous microbiota with 17 rationally selected strains of Clostridia known to produce high amounts of butyrate[17,34] can decrease the severity of GVHD and improve survival across multiple institutions with strain-independent results.

DISCUSSION

The community structure of the microbiota is altered after allo-BMT[4,5]. Our results now provide a novel perspective on microbial metabolites and their effect on GVHD. Our study showed that butyrate was the only SCFA to decrease significantly in amount after allo-BMT, and that the decrease occurred in the intestinal tissue. Decreased amounts of butyrate in the IECs of allo-BMT recipients led to decreased acetylation of histones, whereas increasing the amount of butyrate via intragastric gavage restored histone H4 acetylation and GVHD. An important observation was the lack of changes in amounts of luminal (stool) butyrate, despite a documented shift in the microbiome species that produce low amounts of butyrate after allo-BMT[4]. We posit that this may be because of decreased butyrate uptake into IECs due to decreased amounts of butyrate transporter. In such a case the overall amount of butyrate would not be significantly decreased in the lumen despite decreased production, because less would be taken into the IECs owing to the shift in the microbiota after allo-BMT.

The reasons for decreased amounts of transporter proteins after allo-BMT are intriguing. Previous reports showed a decrease in the amount of SLC5A8 after alterations in the microbiota[38]. Thus, our findings that amounts of SLC5A8 and GPR43 are decreased in IECs after allo-BMT are consistent with previous reports[4,5]. Furthermore, we demonstrate that exposure of IECs to inflammatory cytokines leads to decreased expression of butyrate transporters. These data suggest that the inflammatory milieu early after allo-BMT decreases expression of the butyrate transporter SLC5A8, leading to decreased amounts of SLC5A8 in IECs and further decreasing transporter expression and butyrate intake in a feedback mechanism.

Administered butyrate was more rapidly metabolized in allo-BMT recipients than in syngeneic BMT recipients, as shown by the greater incorporation of carbon from butyrate into the TCA cycle. These data point to a novel observation on the role of energy requirements of IECs in the context of inflammation and GVHD. Our data collectively provide new insights into the role and interactions of the microbiome-derived metabolite butyrate after allo-BMT.

Previous studies have shown that 17 rationally selected strains of *Clostridia* that produce high amounts of butyrate increase numbers of T_{reg} cells in the intestines[17]. Although we did not observe an increase in the number of T_{reg} cells in the intestine or dependence on donor T_{reg} cells for reduction of GVHD severity after butyrate treatment, we cannot formally rule out the contribution of butyrate-mediated effects on T_{reg} cells (directly or indirectly via macrophages) to decreased GVHD severity when the cells are present in the donor inoculum. Furthermore, it is possible that the beneficial effect of the administration of the 17 Clostridial strains was secondary to factors other than butyrate. Nonetheless, when considered in light of the data demonstrating GVHD protection by direct intragastric administration of exogenous butyrate, the results suggest that butyrate is sufficient for GVHD protection.

Our data suggest that butyrate has direct salutary effects on IECs. Butyrate altered the ratio of the expression of anti-apoptotic to pro-apoptotic molecules and increased the expression of proteins relevant for junctional integrity. Nonetheless, the direct physiological relevance of only one of these mechanisms, apoptosis or junctional integrity, cannot be ascertained from our data. Furthermore, our data do not directly address the effects of butyrate on the various other cells that make up the GI epithelium. Clearly, butyrate and HDACis also have a multitude of effects on immune cells, and our data cannot directly measure their effects on GVHD protection. However, a growing body of evidence indicates that GVHD pathophysiology can be regulated by manipulation of the host response to injury, and not just by suppressing the donor immune system[27,39,40]. Our results extend these observations and suggest that direct modulation of target tissues could be an additional important strategy to decrease GVHD severity without resorting to further global immunosuppression.

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METHODS

Methods and any associated references are available in the online version of the paper.

Accession codes. Metabolomic data, mass spectral analytical parameters and spectral raw data from the study, and metadata have been deposited in the NIH Common Fund's Data Repository and Coordinating Center under Metabolomics Workbench Project ID PR000270.

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

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AUTHOR CONTRIBUTIONS

N.D.M. designed experiments, performed experiments, analyzed data and wrote the paper. A.V.M. and M.K. performed experiments and analyzed data. R.J. and A.H. designed and performed experiments. T.T., K.O.-W., S.-R.W., Y.S., C.R., H.F., J.B., Y.S., C.L. and M.C. performed experiments. T.M.S., V.B.Y. and M.v.d.B. wrote the paper. K.H. provided reagents and wrote the paper. S.P. analyzed data and wrote the paper. P.R. designed experiments, analyzed data and wrote the paper.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Reagents. RPMI medium, penicillin-streptomycin, and sodium pyruvate were from Gibco (Grand Island, NY); FCS was from GemCell (Sacramento, CA); 2-ME was from Sigma (St. Louis, MO); mouse GM-CSF was from Peprotech (Rocky Hill, NJ). All antibodies used for flow cytometry were from eBioscience (San Diego, CA). DMSO and butyrate were from Sigma (St. Louis, MO).

Mice. Female C57BL/6J (H-2^b; CD45.2^+) and BALB/c (H-2^d) mice were purchased from the US National Cancer Institute, and FoxP3.DTR (DREG) (H-2^b) mice and C3H.sw (H-2^b) mice were purchased from The Jackson Laboratory (Bar Harbor, ME). The age of mice used for experiments ranged between 7 and 12 weeks. All animals were cared for under regulations reviewed and approved by the University Committee on Use and Care of Animals of the University of Michigan, based on University Laboratory Animal Medicine guidelines.

Cell isolation and cultures. Primary IECs were obtained from C57BL/6J mice as described previously[41]. Luminal contents of intestine were flushed with Ca^{2+}- and Mg^{2+}-free Hanks' balanced salt solution with 1× HEPES-bicarbonate buffer and 2% fetal bovine serum (CMF solution). Intestine was then minced into 0.5-cm pieces; washed with CMF four times; transferred to a mixture of CMF, FBS and EDTA; and incubated at 37 °C for 60 min (with tubes shaken every 10 min). Supernatant containing IECs was then transferred through a 100-µM cell filter and incubated on ice for 10 min to allow sedimentation. Supernatant was then transferred through a 75-µM cell filter. CD326$^+$ IECs were purified using either flow cytometry or anti-allophycocyanin magnetic microbeads (Miltenyi Biotec Ltd., Auburn, CA) and an autoMACS separator (Miltenyi Biotec).

For viability assay, CD326$^+$ IECs were seeded on gelatin-coated 100-mm culture dishes and treated in the presence or absence of different butyrate concentrations overnight. Cells were then subjected to or withheld from irradiation (6 Gy) and cultured for another 24 h.

Bone marrow transplantation. BMTs were performed as previously described[15,42]. Syngeneic (BALB/c→BALB/c or C57BL/6J→C57BL/6J) and allogeneic (C57BL/6J→BALB/c or C3H.sw→C57BL/6J) BMT recipients received lethal irradiation. On day −1, BALB/c recipients received a total of 800 cGy of irradiation (split dose separated by 3 h), and C57BL/6J mice received a single dose of 1,000 cGy. We magnetically separated donor splenic CD90.2$^+$ T cells using an autoMACS (Miltenyi Biotec; Bergisch Gladbach, Germany) and transferred 0.5×10^6 to 2.5×10^6 T cells to BALB/c and C57BL/6J recipients. We transferred 5×10^6 donor whole or TCD bone marrow cells to all recipients. We monitored survival daily and determined recipient body weight and GVHD clinical scores weekly, as described previously[42]. We carried out histopathologic analysis of the GI tract as described[42]. Animals received vehicle (sterile PBS, 0.0004 g Na^+ per dose) or sodium butyrate (10 mg/kg, 0.00004 g Na^+ per dose) by flexible 20-gauge, 1.5-in. intragastric gavage needle daily for 1 week and every other day thereafter.

For BMTs performed at Memorial Sloan Kettering Cancer Center, C57BL/6J mice were treated with an antibiotic cocktail to target obligate anaerobes (5 mg of ampicillin, 4 mg of metronidazole, 5 mg of clindamycin, and 5 mg of vancomycin) that was administered by gavage daily for 6 d (BMT days −18 to −13) and followed 4 and 6 d later by oral gavage with bacteria (BMT days −9 and −7) or PBS. Clostridial bacteria were cultured individually on plates and resuspended in anaerobic PBS at a final OD_{600} of 0.02–0.06. One week later, mice were irradiated (12 Gy, single dose) and transplanted with B10.BR bone marrow cells (5×10^6) and T cells (0.5×10^6 CD5 MACS), and their condition was monitored for development of clinical GVHD.

GC-MS. To determine targeted fatty acid quantitation, we collected samples (plasma, spleen, liver, intestine and intestinal fecal content) from mice 7 d and 21 d after transplant, homogenized them and snap-froze them in liquid N_2. Equal volumes of plasma and homogenized tissue were used, and fecal content was weighed at necropsy. Samples were dispersed in acidified water spiked with stable isotope–labeled SCFA standards and extracted with diethyl ether. We analyzed the ether layer immediately by GC-MS using a Phenomenex ZB-WAX column on an Agilent 6890 GC with a 5973MS detector. Quantitation was done by calibration to internal standards. The tissue levels were normalized by the protein concentration of the homogenized tissue. Heat map data were generated using GenePattern software from the Broad Institute (Cambridge, MA).

Metabolic flux analysis assessing label incorporation into luminal and intestinal tissue butyrate pools. Mice were intragastrically gavaged with a bolus (2 g/kg) of either labeled $^{13}C_2$-butyrate or nonlabeled ^{12}C-butyrate. The small and large intestines were collected 6 h later and prepared for analysis as described above. The incorporation of $^{13}C_2$-labeled butyrate (sodium butyrate-1,2-$^{13}C_2$) in the butyrate pools in the lumen and the intestinal tissue was measured by GC-MS as described previously[43,44], and unlabeled (m/z 145) and labeled (m/z 147) butyrate were detected. The ratio of the labeled to unlabeled peak areas was adjusted to incorporate natural distribution of ^{13}C label and expressed as a percentage of ^{13}C incorporation into the butyrate pool in the luminal contents and intestinal tissue.

Metabolic flux analysis assessing label incorporation into TCA metabolite pools. The incorporation of $^{13}C_2$-labeled butyrate (sodium butyrate-1,2-$^{13}C_2$; Sigma) into the TCA cycle metabolites in the intestinal tissue was measured by liquid chromatography–mass spectrometry performed on an Agilent 6520 high-resolution Q-TOF (quadrupole–time of flight) instrument coupled with an Agilent 1200 HPLC system (Agilent Technologies, New Castle, DE), equipped with an electrospray source. The extract was subjected to hydrophilic interaction chromatography with a Phenomenex Luna NH_2 column (particle size, 3 µm; 1 × 150 mm) at a flow rate of 0.07 mL/min. Solvent A was 5 mM ammonium acetate, pH 9.9, and solvent B was acetonitrile. The column was equilibrated with 80% solvent B. The gradient was as follows: 20–100% solvent A over 15 min; 100% solvent A over 5 min; 20% solvent B for 0.1 min; and 20% solvent A for 15.9 min. Liquid chromatography–electrospray ionization mass spectrometry in the negative mode was done on a Q-TOF instrument with the following parameters: spray voltage, 3,000 V; drying gas flow, 10 l/min; drying gas temperature, 350 °C; and nebulizer pressure, 20 psi. The fragmentor voltage was 150 V in full scan mode. We scanned a mass range of m/z 100–1,500 to obtain full scan mass spectra. We used two reference masses at m/z 121.050873 and m/z 922.009798 to obtain accurate mass measurements within 5 ppm. We extracted and deconvoluted all chromatograms and corresponding spectra of TCA metabolites (citrate, succinate and malate) and their ^{13}C-labeled counterparts using the MassHunter software (Agilent Technologies, New Castle, DE). We rechecked the retention time consistency manually and compared it to that of authentic compounds that were injected under similar chromatographic conditions. For tissue extracts, we normalized the metabolite concentrations to the protein content, which was determined by the Bradford-Lowry method. For the flux analyses, peak areas of the labeled compounds were normalized to the natural abundance of the label and represented as ratios to the total compound peak area.

16S deep sequencing. We collected fecal pellets from mice on specified days (given in figures) and stored them at −80 °C for 3 d prior to analysis. DNA was extracted and purified with phenol-chloroform after bead-based lysis. We carried out sequencing and analysis as described[45]. We sequenced the V4 region of the 16S rRNA gene using Illumina MiSeq technology. We trimmed and analyzed sequencers using mothur[35]. We added the 16S rRNA gene sequence from each strain in the 17-strain cocktail to the version 9 trainset sequences from the Ribosomal Database Project[46]. We classified the resulting sequences by comparing them to the described trainset with the requirement that the confidence score be 100%.

We analyzed Memorial Sloan Kettering Cancer Center 16S experiments using the Illumina MiSeq platform to sequence the V4–V5 region of the 16S rRNA gene. Sequence data were compiled and processed with mothur version 1.34 (ref. 35), screened and filtered for quality[36], and then classified to the species level[47] according to the Greengenes reference database[37].

17-strain mixture. The 17-strain mix was prepared as described[17]. All strains were grown in 5 ml of EG media for 24 h at 37 °C under anaerobic conditions. Each strain was grown to confluence, with the exception of strains 3, 8, 13, 26 and 29. Cells were scraped from EG agar plates and added to the 5-ml culture to obtain the same approximate optical density as the other strains.

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All cultures were then mixed and glycerol was added to a final concentration of 20%. Aliquots (1 ml) were individually frozen and stored at −80 °C for up to 1 year.

Transmission electron microscopy. Samples were stained as described[25]. The intestines from mice that had received butyrate or vehicle treatment were harvested 7 d and 21 d after syngeneic or allogeneic BMT and flushed with 0.1 M Sorensen's phosphate buffer (pH 7.4) with a 20-gauge needle to remove luminal contents. The intestines were next gently flushed with 0.1% ruthenium red (RR) containing 2.5% glutaraldehyde fixative in Sorensen's buffer and immediately placed in a dish containing the stain and fixative. Cross-sections (2 mm wide) were immediately sliced from the duodenum, jejunum and ileum. The tissue was then rinsed three times with Sorensen's buffer containing 0.1% RR and post-fixed for 1 h in 1% osmium tetroxide in the same buffer containing RR. The samples were again rinsed with Sorensen's buffer containing RR. Next, the tissue was dehydrated in ascending concentrations of ethanol, treated with propylene oxide, and embedded in Epon epoxy resin. Semi-thin sections were stained with toluidine blue for tissue identification. Selected regions of interest were ultra-thin sectioned (70 nm thick), mounted on copper grids, and post-stained with uranyl acetate and lead citrate. The samples were examined with a Philips CM100 electron microscope at 60 kV. Images were recorded digitally with a Hamamatsu ORCA-HR digital camera system operated by AMT software (Advanced Microscopy Techniques Corp., Danvers, MA).

FITC-dextran assay. Food and water were withheld from all mice for 4 h on day +21. FITC-dextran (Sigma-Aldrich, St. Louis, MO) was administered by 20-gauge 1.5-inch flexible intragastric gavage needle (Braintree Scientific, Braintree, MA) at a concentration of 50 mg/ml in PBS. BMT recipients received 800 mg/kg (~16 mg/mouse). Four hours later, serum was collected from peripheral blood, diluted 1:1 with PBS, and analyzed on a plate reader at an excitation/emission wavelength of 485 nm/535 nm. Concentrations of FITC-dextran experimental samples were determined on the basis of a standard curve.

Immunoblotting. CD326[+] purified IECs were collected from animals that had received syngeneic (BALB/c→BALB/c) or allogeneic (C57BL/6J→BALB/c) BMT. Whole-cell lysates were obtained, and protein concentrations were determined with the Pierce BCA protein assay (Thermo Scientific). Equal amounts of protein (20 µg) were separated by SDS-PAGE (120 V, 1.5 h) and subsequently transferred to PVDF (polyvinylidene difluoride) membrane via a Bio-Rad semi-dry transfer cell (20 V, 1 h). Antibodies to the following proteins were used to analyze the membranes in dilutions in accordance with the respective manufacturer's specification sheet: α-tubulin (Cell Signaling, clone 11H10), acetyl histone H4 (Lys5/8/12/16) (EMD Millipore, clone 3HH-4C10), SLC5A8 (Abcam), GPR43 (Abcam), occludin (Abcam, clone EPR8208), JAM (Abcam, clone EP1042Y), and claudin 5 (Abcam, ab15106). Secondary anti-rabbit conjugated to horseradish peroxidase (Jackson ImmunoResearch, 111-035-003) was used to detect primary antibodies where needed. Densitometric analysis was done with ImageJ software (US National Institutes of Health, Bethesda, MA).

Quantitative PCR. We isolated mRNA from samples using an RNeasy kit according to the manufacturer's instructions (Qiagen, Venlo, the Netherlands). Using 1 µg of each mRNA template, we synthesized cDNA using SuperScript VILO (Invitrogen, Carlsbad, CA). qPCR primers were designed for murine targets (**Supplementary Table 2**). All primers were verified for the production of a single specific PCR product with a melting curve program.

Chromatin immunoprecipitation. We seeded primary CD326[+] IECs on gelatin-coated (Cell Biologics, Chicago, IL) cell culture dishes (100 mm) overnight and then treated them with 1 mM butyrate for 24 h. We collected the cells and used them for ChIP analysis using the EZ-Magna ChIP kit from EMD Millipore (Billerica, MA) according to the manufacturer's instructions. We cross-linked the cells with 1% formaldehyde and sonicated extracted chromatin using a Bioruptor Pico (Diagenode, Denville, NJ) to yield DNA fragments predominately in the range of 200–1,000 bp. We immunoprecipitated the sonicated lysates using ChIP-grade specific antibodies purchased from EMD Millipore for acetylated histone-H4 and RNA polymerase II or IgG control antibody. Next we examined de–cross-linked DNA by qPCR using primers targeting the promoter regions of the target genes *Bcl2l10* (forward, CCTACTCTGCCTGGCTCTTT; reverse, ACCCTTCTGAGTCCCTGAGA), *Slc5a8* (forward, CACAGCACAG CCTTCTTTGT; reverse, TCCAGTTCACAGTCCAGGTC) and *F11r* (forward, TGCCGGGATTAAAAGCATGG; reverse, ACAGGGACAGCAGGATTAGG). We verified the immunoprecipitation efficiency of all samples by qPCR analysis of the promoter region of *Gapdh* (forward, CTGCAGTACTGTGGGGAGGT; reverse, CAAAGGCGGAGTTACCAGAG). Data analysis was determined as a percentage of input using the following equations: $C_{\mathrm{t[normalized\ ChIP]}} = (C_{\mathrm{t[ChIP]}} - (C_{\mathrm{t[Input]}} - \log_2(6.644)))$ and % input $= 2^{(-\Delta C_{\mathrm{t}}\ [\mathrm{normalized\ ChIP}])}$.

Flow cytometry. To analyze immunophenotype surface markers, we collected lymphocytes contained in the IEC fraction or spleen, stained them using the recommended dilutions indicated on the manufacturer product sheets, and gated them on anti-CD4 conjugated with PerCP/Cy5.5 (clone GK1.5) or anti-CD8 conjugated with allophycocyanin (clone 53-6.7) and configurations of the following per mouse in duplicate: anti-CD69-phycoerythrin (clone H1.2F3), anti-CD62L-phycoerythrin (clone MEL-14), anti-CD25-phycoerythrin (clone 3C7), anti-CD44-PerCP/Cy5.5 (clone IM7), anti-CD44-allophycocyanin (clone IM7) and anti-Foxp3-allophycocyanin (clone FJK-16s). Antibodies to CD4, CD8, CD69, CD62L, CD25, CD44 and CD326 were purchased from BioLegend. Stained cells were then analyzed with an Accuri C6 flow cytometer (BD Biosciences).

IECs were stained with CD326-allophycocyanin (clone G8.8) and DAPI and sorted to >98% purity with a FACSAria III (BD Biosciences) gating on live cells.

All antibodies used have been validated for this species and application as shown in the respective 1DegreeBio validation profile. Antibodies to CD4, CD8, CD69, CD62L, CD25, CD44 and CD326 were purchased from BioLegend, anti-Foxp3 was from eBioscience, and DAPI was from Life Technologies.

Cytotoxic T lymphocyte assay. We isolated CD8[+] T cells from BALB/c (H-2[d]) mice using anti-CD8 microbeads and LS columns (Miltenyi Biotec) according to the manufacturer's instructions. We primed CD8[+] T cells in the presence of irradiated (30 Gy) splenocytes from C57BL/6J (H-2[b]) mice for 6 d before culturing them with primary IECs for 6 h and 16 h. We collected primary IECs from C57BL/6J mice and incubated them overnight in the presence or absence of 1 mM butyrate in gelatin-coated (Cell Biologics, Chicago, IL) 100-mm culture dishes (Fisher Scientific).

Reproducibility. Experiments were repeated at least two times with three sample replicates, bringing the *n* to at least 6; sample sizes are indicated in figure legends. For analysis of one variable in one BMT experiment, at least two groups of three recipients were required.

Statistics. Bars and error bars represent the mean and s.e.m., respectively. We performed non-survival analysis using Student's unpaired *t*-test between two groups. We used ANOVA for comparisons with more than two groups. We performed survival data analysis using a Mantel-Cox log-rank test.

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Corrigendum: Gut microbiome–derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease

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In the version of this article initially published, the middle initial (C) for the sixteenth author was incorrect. The correct author name should be Thomas M. Schmidt (and the corresponding initials in the Author Contributions section should be T.M.S.). The error has been corrected in the HTML and PDF versions of the article.

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